An orbit bound for primitive geometries

A correction to the Rudvalis entry of Leemans–Rodrigues

Leemans and Rodrigues classify rank-two primitive coset geometries using unordered pairs of distinct maximal subgroups, with equivalence under the automorphism group of the ambient group. Table 1 of their paper [1], on physical page 5, gives 1511 geometries for the Rudvalis group. Under precisely this convention, a single conjugacy class of maximal subgroups already gives at least 50669. The argument is an orbit bound; it does not require enumerating these geometries.

Theorem 1. *Let G be a finite nonabelian simple group with trivial outer automorphism group. Suppose that M is a maximal subgroup of order m, and put $n = |G|/m$. There are at least*

$$\left\lceil \frac{n-1}{2m} \right\rceil$$

equivalence classes of firm, connected, primitive rank-two coset geometries whose two parabolic subgroups are conjugate to M. Equivalence here allows interchange of the two types and is induced by an element of $\mathrm{Aut}(G)$.

Proof. Maximality gives $N_G(M) = M$ or G. The latter alternative makes M normal. It is impossible because G is simple and M is a nontrivial proper subgroup. Thus M has exactly n conjugates. Let Ω be this set, and let X be the set of its two-element subsets. Then $|X| = n(n-1)/2$.

Each $\{H, K\} \in X$ defines a rank-two coset geometry. The right cosets of H and K are incident when their intersection is nonempty. Every incident pair is carried to $\{H, K\}$ by right multiplication, so the action is flag-transitive. Its local valencies are $[H : H \cap K]$ and $[K : H \cap K]$. Both exceed one: inclusion between two distinct maximal subgroups is impossible. Hence the geometry is firm. Moreover $\langle H, K \rangle = G$ by maximality. Paths in the incidence graph correspond to alternating products from H and K, so this equality proves connectedness. Maximality of both parabolic subgroups gives primitivity on each type. Simplicity also makes both coset actions faithful, since their kernels are proper normal subgroups.

The equivalence stipulated in the statement is exactly the orbit relation of $\mathrm{Aut}(G)$ on X. Since G is centreless and has trivial outer automorphism group, $\mathrm{Aut}(G) = \mathrm{Inn}(G) \cong G$. Every orbit therefore has at most $|G|$ elements. Dividing $|X|$ by $|G| = nm$, and rounding upwards, proves the bound. The factor two has already allowed interchange of H and K; it must not be charged again. □

Corollary 2. *There are at least* 50669 *primitive rank-two coset geometries for the Rudvalis group, with equivalence as in [1]. In particular, the entry* 1511 *in its Table 1 is false under its stated convention.*

Proof. We retain the following classical group facts from the Atlas [2]: the Rudvalis group is nonabelian simple, has order 145926144000, has trivial outer automorphism group, and contains a maximal subgroup $(5{:}4) \times A_5$. These are inputs, not new results of this note. The subgroup has order $20 \cdot 60 = 1200$, and hence

$$n = \frac{145926144000}{1200} = 121605120.$$

Theorem 1 now gives

$$\left\lceil \frac{121605119}{2400} \right\rceil = 50669 > 1511.$$

Both defining subgroups belong to just this one conjugacy class, so including all maximal subgroups can only increase the number. □

There is also a useful exact formula underlying the bound. It separates the counting problem from the construction of representatives.

Theorem 3. *Let a finite group A act on a finite set Ω, and let $\psi(a) = |\operatorname{Fix}_\Omega(a)|$. The number of A-orbits on two-element subsets is*

$$\frac{1}{2|A|} \sum_{a \in A} \bigl(\psi(a)^2 - 2\psi(a) + \psi(a^2)\bigr). \tag{1}$$

For a nonabelian simple group G, take $A = \operatorname{Aut}(G)$ and Ω to be all maximal subgroups of G. Formula (1) then counts exactly the primitive rank-two geometries with the equivalence of [1].

Proof. A two-element subset fixed by a either consists of two individually fixed points, or forms a two-cycle of a. There are $\binom{\psi(a)}{2}$ subsets of the first kind. Of the $\psi(a^2)$ points fixed by a^2, exactly $\psi(a)$ are already fixed by a; the remainder occur in two-cycles. Thus the number of fixed subsets is $(\psi(a)^2 - 2\psi(a) + \psi(a^2))/2$. Burnside's orbit-counting lemma gives (1). The last assertion follows from the same coset and maximality argument as in Theorem 1. □

The bound in Corollary 2 is the certified numerical conclusion. It is not a corrected exact total or a construction of every geometry. The other eleven rows of Table 1, the printed orbit-length tables, the designs and the external census data require separate verification. No claim about them follows merely from this contradiction. Nor does this note establish that the fixed edition has no later correction.

The source supplies the geometry problem and equivalence convention; the Atlas supplies the stated group data. Orbit–stabilizer and Burnside's lemma are classical. Our contribution here is their application to the source's numerical assertion and the explicit separation between counting orbits and constructing representatives. The accompanying integer certificate checks the displayed arithmetic; the general argument above supplies the proof. We claim neither a newly solved open conjecture nor a compression factor in this note.

References

[1] D. Leemans and B. G. Rodrigues, *On primitive geometries of rank two*, The Art of Discrete and Applied Mathematics **2** (2019), P1.08. doi:10.26493/2590-9770.1301.165.

[2] R. A. Wilson, R. A. Parker and J. N. Bray, *Atlas of Finite Group Representations: Rudvalis group Ru*, group data and maximal subgroup list, accessed 14 October 2026. `https://brauer.maths.qmul.ac.uk/Atlas/spor/Ru/`.